Iterative Time Complexity

1. **Past Question -** CSE221_SolutionOfPastQuestions_TimeComplexity [Student View]
2. **Open DSA -** 4.6. Calculating Program Running Time — CS2 Software Design & Data Structures
3. **Princeton -** COS226 Analysis of Algorithms: Practice Examples Spring '20
4. **UNSW -** COMP 9024, Homework Solution, 11s2, Class 1
5. **UMBC -** CMSC 441: Homework #1 Solutions
6. **NUS -** Tutorial 9 – Analysis of Algorithms 1. Big-O Complexity
7. **HFN -** 1.3 time complexity Additional Solution and prep.pdf
8. **Examples shown in class** - Iterative Time Complexity Examples
9. **ANK** - Practice sheet time complexity

Recursive Time Complexity

1. **HFN -** 2. Recursion time complexity HFN.pdf
2. **Master Theorem -** master.pdf
3. **Examples shown in class** - Recursive Time Complexity Examples

Divide & Conquer

1.

Renowned Progressive Rock band 'Porcupine Tree' released an album called 'Closure/Continuation' after about 13 years. Now as a Progressive Rock Music fan you are going to listen to the tracks of the album but in the order of their Youtube views (highest one at first, lowest one at last) instead of the order of the album tracklist.

You have chosen an Algorithm to order them as per your preference. If multiple tracks have the same views, you are going to listen to any of them the Algorithm puts first in the list after ordering. This algorithm also solves your concern about the issues regarding space that your device is facing. Porcupine Tree made the fans wait for 13 years for a new album and so you think you have the patience to wait as much time as the Algorithm may take to order the tracks.

The following table contains the list of the tracks and their Youtube views (in thousand) :

Track	Harridan	Of The New Day	Rats Return	Dignity	Herd Culling	Walk the Plank	Chimera's Wreck	Population Three
Views	15	8	11	112	33	39	88	41

a. CO2 **Specify** the name of the algorithm you have chosen and **simulate** the Algorithm to order the tracks of the album as per your preference. Show your workings in detail. **7**

b. CO5 Suppose you want to run the algorithm again on the ordered list. **Determine** the run-time complexity of the algorithm in this scenario. **3**

2.

Jack loves to play with integers. He created a list of n integers where the even indices hold numbers in decreasing order and the odd indices hold numbers in increasing order. For example, this is a list of n=8 integers Jack made.

Index	0	1	2	3	4	5	6	7
Number	23	2	19	3	7	11	5	13

[Explanation:
The indices 1, 3, 5, and 7 have numbers 2, 3, 11, and 13 in increasing order.
The indices 0, 2, 4, and 6 have numbers 23, 19, 7 and 5 in decreasing order.]

You sorted the the list in quadratic time [$O(n^2)$]. To your utter surprise, Jack replies, "It could be sorted in linear time".

1) ***[CO2]*** **Show** the steps how you sorted the list using a suitable algorithm. Mention the name of the algorithm. — 3
2) ***[CO2]*** **Describe** how Jack sorted the list in linear time. Show the steps too.— 4
3) ***[CO3]*** Jack wants to add the number 15 to your sorted list from (1). He wants you to come up with an efficient idea about how he is going to find the accurate index position for 15 in the list.
   Describe how you are going to assist him in this. **Show** the steps too. — 3

3.

You are given an array containing N distinct integers in a wave-like sequence. Meaning, the numbers in the beginning are in ascending order, and after a specific position, they are in descending order. For example: [1, 3, 4, 5, 9, 6, 2, -1]
You have to find the maximum number of this sequence. Can you devise an efficient algorithm such that the time complexity will be less than $O(N)$?

a) **Present** your solution idea as a pseudocode/ python code/ flowchart/ step-by-step instructions/ logical explanation in one-two paragraphs.
b) **Write** the time complexity of your algorithm.

While sorting a list of 10 integers in ascending order using Quick Sort, after the first partition (using the first element as a pivot), the list looks like this : [13, 11, 19, 7, 23, 37, 29, 53, 59, 41]. Which element was the pivot before partitioning?
Explain your answer in brief.

4.

Suppose the company, "Sparkling Ideas," has been new in the market for the past few years and you work as a financial analyst hired for this company.

You have been provided a list of the company's monthly incomes over the last year. However, the incomes are mixed with positive and negative figures, which indicates the company has faced some difficulties during certain months. The company's CFO has a special skill to identify the trend in the company's income details, which helps her make informed decisions. She believes that there is a particular set of consecutive months where the company earns the most, and she wants you to find out which one(s) it is.

Your task is to find the maximum earnings from the given monthly incomes and determine in which part of the year the company earned the most.

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
7	-15	9	-4	12	-8	3	-11	16	-2	6	-10

The figures are in millions of USD.

a) Try to think of an efficient algorithm to solve this problem and provide the result in terms of the starting and ending months, along with the corresponding total income. **Show** a simulation of your algorithm.
b) **Calculate** the time complexity of your algorithm. Show proper mathematical logic.

5.

You are given an array containing N distinct integers in a wave-like sequence. Meaning, the numbers in the beginning are in descending order, and after a specific position, they are in ascending order. For example: [9, 7, 6, 4, 2, 1, 3, 5, 8]
You have to find the minimum number of this sequence. Can you devise an efficient algorithm such that the time complexity will be less than $O(N)$?

c) **Present** your solution idea as a pseudocode/ python code/ flowchart/ step-by-step instructions/ logical explanation in one-two paragraphs.
d) **Write** the time complexity of your algorithm.

6.

While sorting a list of 10 integers in ascending order using Quick Sort, after the first partition (using the first element as a pivot), the list looks like this : [31, 19, 3, 17, 23, 37, 59, 61, 71, 43].
Which element was the pivot before partitioning?
Explain your answer in brief.

7.

Inspired by Karatsuba's algorithm for multiplying integers, your friend has come up with the following divide-and-conquer algorithm for squaring n-digit numbers. Pseudocode for the algorithm is given below. Your job is to fill in the details. You may assume that n is a power of two.

Algorithm 1: KARAT-SQUARE(x)

```
   Input  : An n-digit number x
   Output: x^2
1  if n = 1 then
2  |  return x^2
3  end
4  a ← the leftmost n/2 digits of x;
5  b ← the rightmost n/2 digits of x;
6  S1 ← KARAT-SQUARE(a) ;                    /* S1 equals a^2 */
7  S2 ← KARAT-SQUARE(b) ;                    /* S2 equals b^2 */
8  S3 ← KARAT-SQUARE (_____) ;               /* fill in the blank */
9  P ← _____ ;                               /* P should equal 2ab */
10 return _____ × 10^n + P × 10^(n/2) + _____;
```

Now carefully read the algorithm above, and answer the following questions.

i) Write x in terms of a and b. Looking at lines 4 and 5 of the algorithm should help.
ii) Square your answer from (i). The formula $(m + n)^2 = m^2 + 2mn + n^2$ should help.
iii) Fill in the blank in line 8. Your answer should involve a and b in some way. The idea is to compute a square that, along with a^2 and b^2, helps you compute $2ab$. You may want to think about questions (iii) and (iv) at the same time.
iv) Fill in the blank in line 9. Your answer should involve S_1, S_2, and S_3 in such a way that P equals $2ab$.
v) Fill in the blank in line 10.
vi) Write the running time of your algorithm above.
vii) Say you have another algorithm KARAT-SQUARE-2(x) that takes an n-digit number and squares it by recursively squaring six $n/3$-digit numbers, and combining them in $O(n)$ time. Which algorithm is asymptotically faster: KARAT-SQUARE(x) or KARAT-SQUARE-2(x)?

8.

Suppose the company, "Sparkling Ideas," has been new in the market for the past few years and you work as a financial analyst hired for this company.

You have been provided a list of the company's monthly incomes over the last year. However, the incomes are mixed with positive and negative figures, which indicates the company has faced some difficulties during certain months. The company's CFO has a special skill to identify the trend in the company's income details, which helps her make informed decisions. She believes that there is a particular set of consecutive months where the company earns the most, and she wants you to find out which one(s) it is.

Your task is to find the maximum earnings from the given monthly incomes and determine in which part of the year the company earned the most.

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
17	-15	9	-4	12	-18	3	11	16	-2	-16	10

The figures are in millions of USD.

c) Try to think of an efficient algorithm to solve this problem and provide the result in terms of the starting and ending months, along with the corresponding total income. **Show** a simulation of your algorithm.
d) **Calculate** the time complexity of your algorithm. Show proper mathematical logic.

9.

Imagine you're a farmer named Sam who grows different types of vegetables in several continuous fields. Each field can grow a specific type of vegetable, and based on the market value of that vegetable, each field will bring a certain profit (positive integer) or loss (negative integer). The profit or loss is estimated and noted for each field.

Sam can start farming from any field, but once he starts, he must continue farming the next fields in sequence without skipping any, until he decides to stop, because his tractor can only move to the next adjacent field and cannot skip fields.

Sam needs to choose which sequence of fields to farm to maximize his profits. The following array represents the estimated profits or losses for ten consecutive fields on Sam's farm, based on the types of vegetables that can be grown in each field and their respective market values.

[4, -18, 16, -14, 12, -1, 3, 1, -20, 15]

a) Can you help Sam decide which sequence of fields to farm for maximum profit using an efficient algorithm? What would the profit amount be? **Show** a simulation of your proposed algorithm. You must show which sequence of fields he needs to select in order to achieve this maximum profit.
b) **Calculate** the time complexity of your algorithm using proper mathematical logic. An efficient algorithm should have time complexity less than or equal to $O(N * log(N))$ where N is the number of fields.

10.

Simulate the Quick Sort algorithm by showing the steps on the following array, always choosing the last element as the pivot element. You can partition the array at each step choosing any suitable algorithm, *but you do not need to show the partitioning steps*.

51	70	28	12	75	62	88	43

11.

John is an amateur poker player who has decided to try his luck at a local casino. The casino allows players to play in rounds, each leading to a win (positive integer) or a loss (negative integer). John keeps track of his net gain or loss after each round.

John knows that poker isn't just about luck—it's about making strategic decisions. And one of the critical decisions he has to make is knowing when to stop playing. However, the casino has a rule that once a player starts playing rounds, they must play the subsequent rounds in order without skipping any, until they decide to stop, as skipping rounds might disrupt the flow of the game and disadvantage other players.

With this in mind, John wants to find the optimal sequence of rounds where he can maximize his net winnings. The following array represents the net gain or loss John experiences in ten rounds of poker at the casino.

[8, -18, -8, 2, 16, -14, 2, -1, 3, 15]

c) Find an efficient algorithm that will help John to figure out which sequence of rounds he should play to maximize his winnings. What would be the exact profit amount? **Show** a simulation of your proposed algorithm. You must show which sequence of rounds he needs to select in order to achieve this maximum profit.
d) **Calculate** the time complexity of your algorithm using proper mathematical logic. An efficient algorithm should have time complexity less than or equal to $O(N * log(N))$ where N is the number of fields.

12.

Consider an array containing **N** unique values where for some index **i**, the values are in increasing order from index **0** to (**i-1**), and then again from **i** to **(N-1)**. An example array is given below.
Here **i=3**, it means the values are in increasing order from index 0 to 2, and then again from 3 to 7.

index	0	1	2	3	4	5	6	7
value	9	12	15	2	4	5	7	8

Given the value of **i**, propose an algorithm to search a *key_value* in the array. Complexity of your algorithm must be less than **O(N)**.

a) Present your solution idea as a code/ pseudocode/ flowchart/ step-by-step instructions/ logical explanations in short.
b) Write the time complexity of your presented solution.
c) Show a simulation of the Merge Sort algorithm to organize the whole array in increasing order.

13.

Your friend gave you a binary string B (meaning each character is either **0** or **1**). He wanted to find out how to calculate the maximum number of consecutive 0s in that particular string. For example,

String: 100100000111	Maximum consecutive 0s: 5
String: 1010101010101	Maximum consecutive 0s: 1

You, as an algorithm enthusiast, know that this can be solved in linear time. However, your friend asked you to propose a Divide and Conquer approach

a) Name a suitable Divide and Conquer algorithm for this task.
b) Explain how you can apply that algorithm in this scenario. Present your idea in a pseudocode/programmable code/Flowchart/step-by-step instructions.
c) Write the time complexity of your algorithm.

Graph

1.

Bill has this weird characteristic of playing with different types of graphs. This time he comes up with an undirected simple graph of 10 nodes. Nodes are labeled from 3 to 12. Some of the edges are:
(5,6), (5,10), (6,11), (7,11). He created other edges following two rules.

a. All the nodes that are labeled with a number which is a multiple of 3 have edges among them.
b. All the nodes that are labeled with a number which is a multiple of 4 have edges among them.

a. CO3	**Draw** the graph to show all the edges.	**1**
b. CO3	Bill says, there are at least four triangles in the graph. Do you agree? **Support** your answer by showing the nodes which form these triangles.	**2**
c. CO2	**Simulate** a BFS algorithm on the graph to find the shortest distance from Node 3 to all others nodes.	**6**
d. CO3	Bill told you to keep on adding edges between the nodes according to your wish, keeping the graph simple (without adding multiple edges between any two nodes, self edges). **Compute** the number of more edges that you can add.	**1**

Bill has this weird characteristic of playing with different types of graphs. This time he comes up with an undirected simple graph of 10 nodes. Nodes are labeled from 3 to 12. Some of the edges are:
(5,6), (5,10), (6,11), (7,11). He created other edges following two rules.

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b. All the nodes that are labeled with a number which is a multiple of 4 have edges among them.

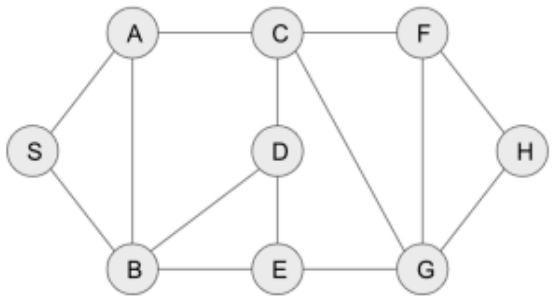
a. CO3	**Draw** the graph to show all the edges.	**1**
b. CO3	Bill says, there are at least four triangles in the graph. Do you agree? **Support** your answer by showing the nodes which form these triangles.	**2**
c. CO2	**Simulate** a BFS algorithm on the graph to find the shortest distance from Node 3 to all others nodes.	**6**
d. CO3	Bill told you to keep on adding edges between the nodes according to your wish, keeping the graph simple (without adding multiple edges between any two nodes, self edges). **Compute** the number of more edges that you can add.	**1**

2.

3.

Total Marks: 10**[CO3]**

Consider this graph:



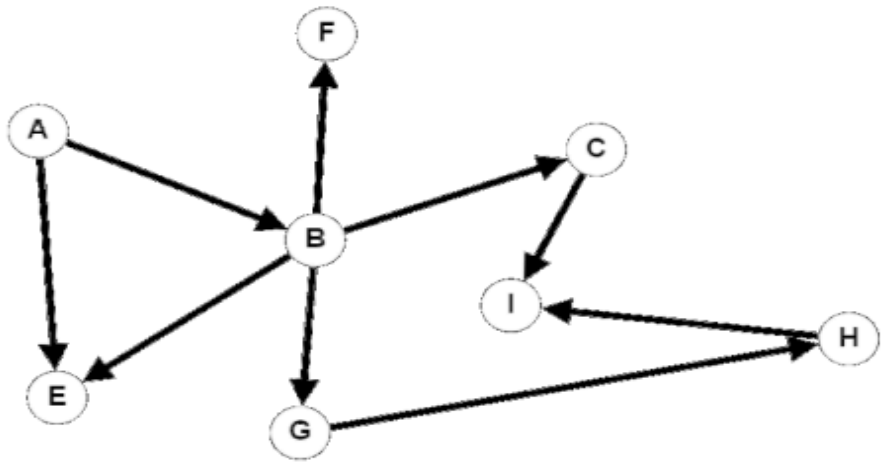
a. **Justify** $\sum_{v \in V} deg(v) = 2m$, where m = number of edges in the graph. 2

b. **Create** an adjacency matrix for this graph.

c. **Apply** Depth First Search to traverse this graph considering S to be the source vertex. **Demonstrate** the results. 2

6

4.



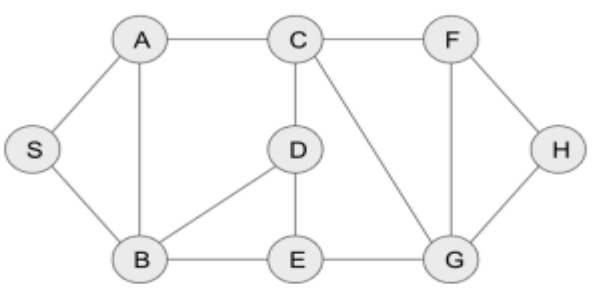
(a) **Apply** the DFS algorithm on the following graph starting from vertex **A**. Determine start time, finish time, and parent for each of the vertices. Choose the smallest (in alphabetical order) vertex when there is a choice. **06**

(b) For each edge, **determine** whether it is a tree edge, forward edge, back edge or cross edge. **04**

5.

Total Marks:10**[CO3]**

Consider this graph:



a. **Justify** $\sum_{v \in V} deg(v) = 2m$, where m = number of edges in the graph. 2

b. **Create** an adjacency list for this graph.

c. **Apply** Breadth First Search to traverse this graph considering S to be the source vertex. **Demonstrate** the results. 2

6

6.

The cities in Italy are connected in such a way that their adjacency list representation looks as follows-

Rome: Milan, Venice, Florence
Milan: Rome, Florence, Bologna, Venice
Florence: Milan, Naples, Rome, Bologna, Venice
Venice: Rome, Milan, Florence
Naples: Florence, Bologna, Genoa
Turin: Genoa
Bologna: Naples, Genoa, Milan, Florence
Genoa: Turin, Bologna, Naples
Palermo:Catania, Syracuse
Catania: Palermo, Syracuse
Syracuse: Catania, Palermo

Suppose Rome suddenly gets affected with covid-19. The rest of the cities are non-affected. You are asked to find out the time steps the covid virus will take to affect all the non-affected cities if it takes one time step to travel from one city to another. So, one affected city will affect all of its neighbors in one time step.

a) **Draw** the graph and create the adjacency matrix for the graph. **04**
b) **Mention** the name of the graph traversal algorithm you think is suitable for solving the problem. **01**
c) **Simulate** the suitable algorithm to find the time steps needed for each non affected city to be affected. Choose the smallest (in alphabetical order) vertex when there is a choice. **05**

7.

a. **CO1** You have the following adjacency matrix for a graph. However, some of the entries are missing. Your job is to find these missing entries with the help of some clues. Then draw the graph. **06**

	A	B	C	D	E	F
A	0	0		0	0	0
B	1	0	1	0	0	1
C	0	0	0	0	1	0
D	0	0	1	0	0	0
E	0		0	0	0	
F	1	0	1	0		0

Clues:
- B can not be reached from C
- A can be reached from D
- 2 x |E| = 3 x |V| (twice the no. of edges is equal to thrice the no. of vertices)
- In-degree of C is not a prime number.

b. **CO2** Is an adjacency list more memory efficient than an adjacency matrix? Justify your reasoning with respect to directed vs undirected, weighted vs unweighted, sparse vs dense graphs. **04**

8.

Dhaka, especially your area, is facing an Electricity crisis. Many initiatives including load shedding, reducing office hours, etc have already been taken. The local authority of your area is looking for feasible solutions to reduce the consumption of electricity. One of the members of their advisory committee has suggested that they should reduce the operating costs of road lighting. Till now every road is illuminated all night long. To reduce electricity consumption, they have decided to no longer illuminate every road, but to switch off the road lighting of some roads. To make sure that the inhabitants of your area still feel safe, they want to optimize the lighting in such a way that after darkening some roads at night, there will still be at least one illuminated path from every major point in your area to every other major point.

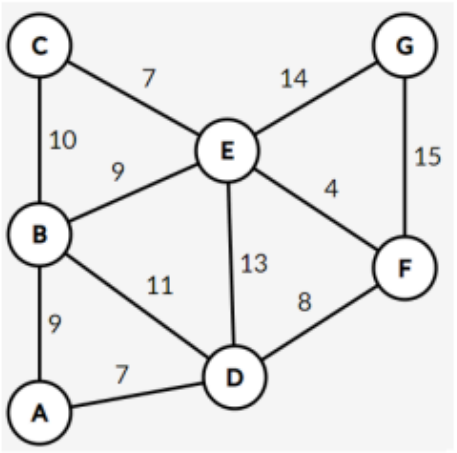


Figure : Graph of your area where the Vertices denote major points and Edges denote electricity consumption in a day.

a. CO3 **Apply** a suitable Algorithm to help the authority determine the roads they need to illuminate and the cost which will minimize the electricity consumption. If you need to consider any root vertex for your Algorithm, you can consider 'A' as such.

b. CO5 Observe the Graph of your area and your provided solution of question 1(a) and **determine** the maximum electricity of a day the authority can save.

c. CO6 The United Arab Emirates (UAE), especially Dubai, is hosting the Asia Cup 2022. They too want to minimize the electricity consumption by switching off the lights in some roads leaving exactly one illuminated path from every major point in your area to every other major point. However, to keep the tourists entertained, the Authority of Dubai has taken an opposite approach to yours. They want to illuminate the roads which require the most electricity.
Propose an algorithm to compute the path which fulfills this requirement.

a. **CO3** As we all know, the Spanish National Football Team is notorious for their passing football, especially their back passes. After their disastrous 2022 World Cup campaign, The King of Spain made you the coach and declared that no back passes are allowed on the field from now on till Spain scores 2 goals, that is, Simon the goalkeeper can only pass to the defenders Rodri and Cesar, but Rodri and Cesar cannot pass the ball back to Simon. Also, every pass now comes with a cost– and the more cost players accumulate, the angrier the King gets. These are the costs of the passes:

Passes	Cost
Simon → Rodri	2
Simon → Cesar	4
Rodri → Cesar	1
Rodri → Pedri	7
Cesar → Gavi	3
Gavi → Pedri	2
Pedri → Morata	1
Gavi → Morata	5

Find the lowest cost for the ball to reach every player starting from Simon, using any algorithm you find useful. The King will need to see every single step of the algorithm, or he will not believe you and fire you on the spot.

9.

b. **CO3** After scoring 2 goals, Spain changed their strategy and started to back pass again incorporating 2 more players, the master of dark arts Busquets and the young sensation Ansu Fati. Now they are no longer concerned with the cost of the pass as well. These are the passes for the new strategy:

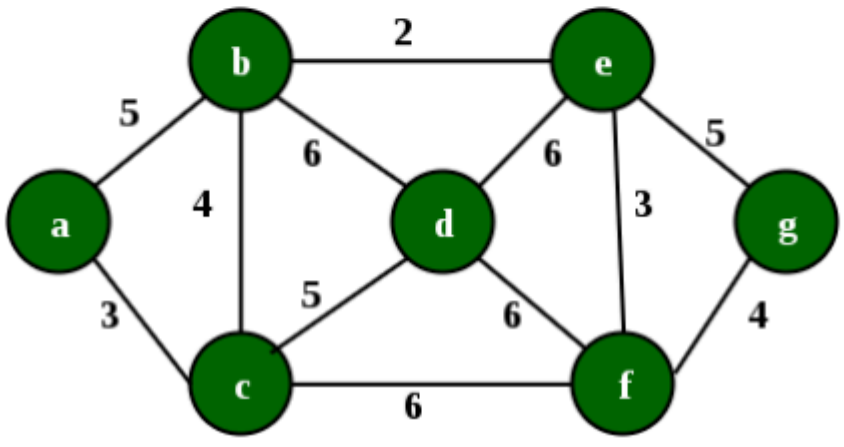
Passes
Simon → Rodri
Simon → Cesar
Rodri → Busquets
Rodri → Pedri
Cesar → Rodri
Busquets → Gavi
Busquets → Pedri
Gavi → Cesar
Gavi → Pedri
Gavi → Fati
Fati → Morata
Pedri → Fati
Morata → Pedri

Now using a suitable algorithm, **find** out the largest group of players who can pass the ball among themselves. For example, one such group can be (Fati, Morata, Pedri) where Morata can pass to Pedri, Pedri can pass to Fati and Fati can pass to Morata. It is important to keep in mind that the king will be observing every step of the algorithm. Any discrepancies can lead you to lose your job.

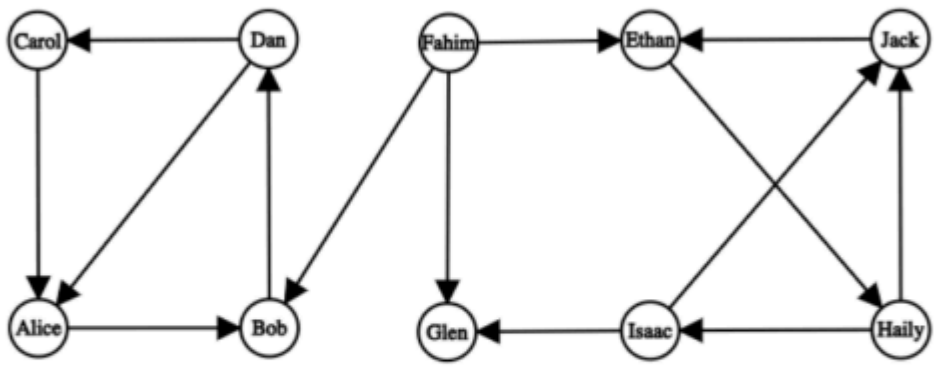
10. You work as an engineer for the Roads and Highways department in a district. The district has 7 thanas that are represented by the 7 vertices in the following graph. The edges between the vertices represent the roads that connect one thana with another and the weight of an edge represents the length of the road.

A recent flood has totally damaged these roads and immediate repair work is needed. The cost to repair a road is proportionate to its length. However, your department does not have the budget to repair all the roads so you need to repair a subset of the roads. Which algorithm should you use here to find out the roads that need to be repaired to keep all the thanas connected to one another with the **minimum** possible cost?

Show the step by step simulation of the algorithm. You may pick any algorithm of your preference that you think will solve the problem.

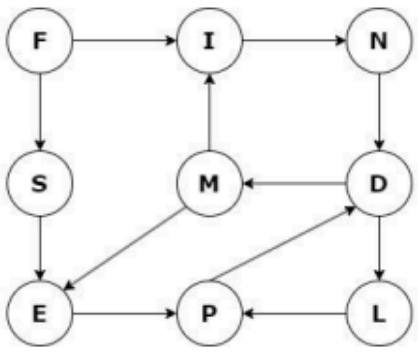


11. Among a group of 10 friends, it's not feasible for each individual to visit the houses of all the others. With an upcoming tournament in mind, the friends need to establish groups. The criterion for forming these groups is that a friend must be capable of visiting the houses of every other member in their group. The task involves calculating the count of viable groups following this criterion. A group has to consist of a minimum of two people.



a) [CO1] The end result will showcase the list of groups and the members. **Name** the algorithm tailored for this situation.
b) [CO1] **Show** a simulation of your chosen algorithm using the graph above. **Write** the number of groups that can be formed and the members of each group.
c) [CO1] Those who are incapable of forming a group will be identified as individuals ineligible for tournament participation. **Identify** those people if there are any.
d) [CO3] We also want to find the person who can visit the most houses. **Propose** an algorithm to implement this. Present your algorithm with a pseudocode/flowchart/step-by-step instructions.

1. Sherlock Holmes found himself in a dire situation where James Moriarty imprisoned his assistant Watson in a maze. The maze is made of 9 different rooms. Each room has a part of the key with an alphabet written on it. Moriarty tells Sherlock that only if he can attach all the key chain parts in exactly this order: **"F-I-N-D-M-E-P-L-S"**, then he can unlock the room with that key where Watson is kept. The map of the maze is given below:



a) [CO1] Now you know something about a graph traversal algorithm that travels from the source to as far as possible along each branch (depth-wise traversal). Simulate/Draw the traversal and **fill up** the following table (in your script) with the start and ending time for each node you achieved while doing this task. Keep in mind, wherever you have to choose between different neighbors to explore, follow the sequence that will make the key chain parts in the sequence that Sherlock needs.

Node	D	E	F	I	M	S	L	P	N
Start									
End									

b) [CO1] When Sherlock tries to save his friend, water suddenly fills the rooms. He needs to find a way to contain the water somehow so that the water doesn't overflow from one place to another. So, he created different isolated closed groups of rooms where he can travel anywhere in that particular group, meaning they have a path from any room to any other room inside the group. After doing so, he finds out that he can make a few of these groups of rooms inside this maze. A group can contain 1 to N number of rooms where N is the total number of vertices, and no two groups can share the same room.

i) **Mention** the name of a suitable algorithm for this task.

ii) **Show** simulation for the graph given above.

You can reuse/refer to your answer from question (a) to reduce work.

12.

[CO2] "Dijkstra's Algorithm can find the single source shortest paths for any graph."
Do you think this statement is true? If yes, **state** your reasoning. Else, **validate** it to be false with an example graph.

13.

14.

You and your friend learned about single-source shortest path algorithms and are now given a directed weighted graph (*figure 2.*) to implement your algorithm. However, **the edge (2,4)** has an undefined weight **(Weight: X)**.

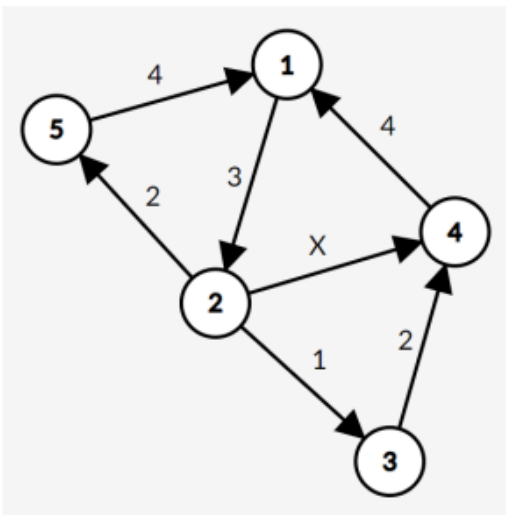


Figure 2: A Directed Weighted Graph for Question 2

a. **CO2** If the edge weight **X = 1**, then simulate the Dijkstra Algorithm with the necessary calculation needed to find the shortest path from **Source Node 1** to every node in the graph. Mention the minimum distances you could find. **03**

b. **CO1** Now, with the weight of edge (2,4) set to **X = -9**, prove or disprove the statement: '*This graph's single-source shortest path is unsolvable from source 1, and only Bellman-Ford can identify this issue.*' Simulate/Give logical reasoning to support your answer. **04**

c. **CO1** If all roads have the same travel time, can an algorithm with time complexity O(V+E) be used to find the best route? Explain your idea in brief statements/pseudocode/flow-chart/step-by-step instructions. **03**

15.

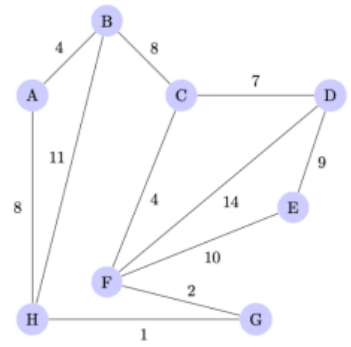


Figure 1: A Transportation Network

a. For the given transportation network, find a configuration connecting all regions with minimal cost. **Demonstrate** the steps and the algorithm you used to select connections.

b. Based on the result from part (a), **calculate** the total cost of your configuration and explain in short, why it represents the minimum cost required to connect all regions.

c. Now, imagine the road between Region C and Region F became ***unavailable*** due to maintenance. How would this impact overall connectivity? Propose an alternative solution to maintain network connectivity with minimal disruption, and explain your reasoning.

16.

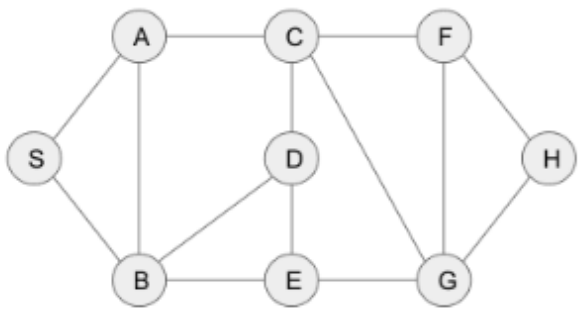
Problem #3:

The following information is stored after implementing the DFS algorithm on an undirected graph.

Nodes	1	2	3	4	5	6	7
Parent							
Starting Time	8	11	5	6	4	1	2
Finish Time	9	12	10	7	13	14	3
Distance from Root							

A. Draw the DFS tree.
B. Fill up the empty rows of the table.

17.



A. Justify $\sum_{v \in V} deg(v) = 2m$, where m = number of edges in the graph.
B. Calculate the number of additional edges that can be added between the nodes randomly, keeping the graph 'simple' (without adding multiple edges between any two nodes, loops in a single node).

18.

Problem #6:

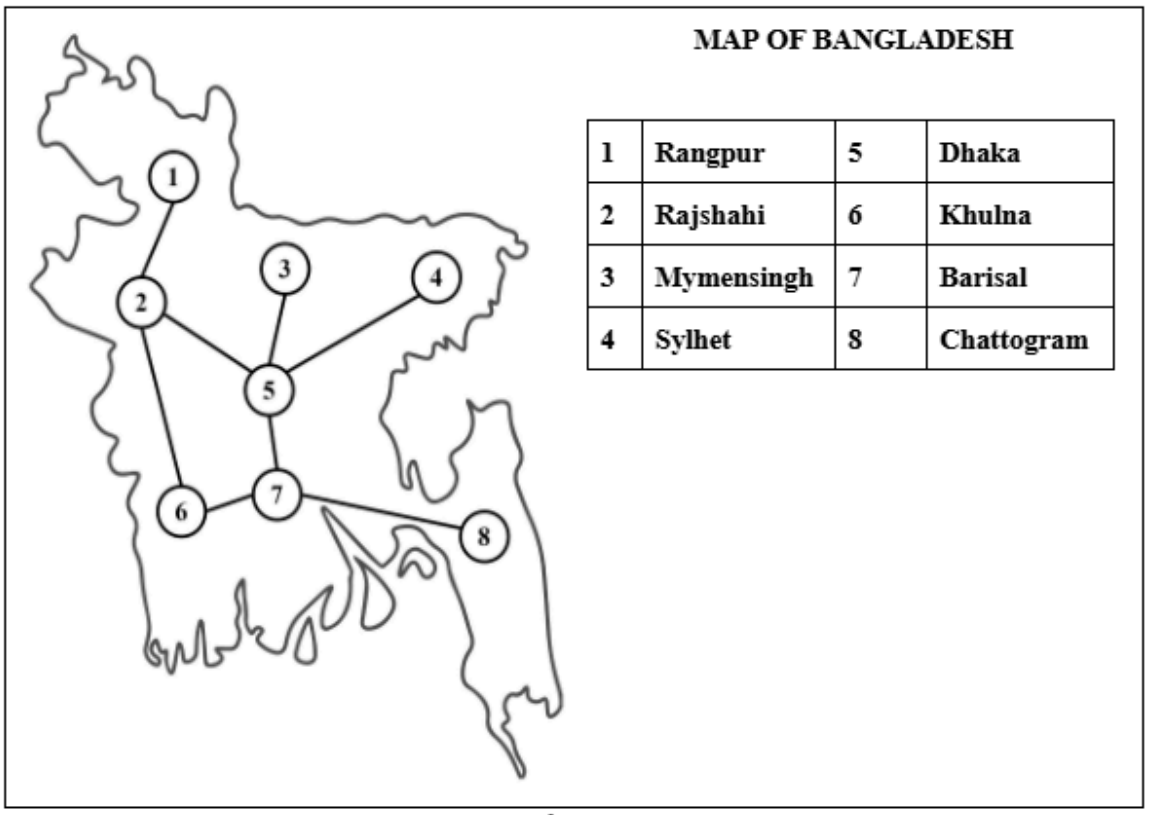
Suppose you are given a directed graph where the number of vertices is V and the number of edges is E. The vertices are numbered from 1 to V. You are given the task to find out whether vertex '1' is reachable from all the other vertices, i.e. there exist paths to go from '2' to '1', '3' to '1', … and so on.

Propose an *O(V+E)* algorithm to find the answer to this question.

Clarification: You cannot run DFS from each source. That would take the complexity to *O(V*E)*.

19.

The "Road Transport and Highways Division" of Bangladesh is working on a project where they want to establish six-lane highways among the 8 divisions of Bangladesh. Currently, there are 8 highways connecting these 8 divisions as given in the map below.



1	**Rangpur**	**5**	**Dhaka**
2	**Rajshahi**	**6**	**Khulna**
3	**Mymensingh**	**7**	**Barisal**
4	**Sylhet**	**8**	**Chattogram**

Fig 1: Graph For Question 4

i. **Create** a graph representation for the given map so that it does not take more than $O(|V| + |E|)$ space/memory.

ii. **Determine** how many additional highways we need to make sure that every division is directly connected with each other.

iii. In a meeting, the director looked at the map and wondered if the divisions of Bangladesh could be split into two groups (**A** and **B**) such that all highways directly connect locations in Group A to Group B, with no highways between locations within the same group. Using an appropriate data structure, simulate the algorithm that can determine if such a division is possible.

iv. Based on the given graph in Fig 1, **explain** how you can find out if there is an odd number cycle present in the graph. An odd number cycle is a cycle that has an odd number of vertices.

Greedy Algorithms

1.

Alice and Bob both are the peers of a decentralized distributed network. Alice wants to send a message to Bob using the platform but she does not want that any other users of the network would be able to understand the message. That is why she encodes the message using an encoding algorithm which is known to only both of them and sends it to Bob. The message is:

Abaaccdd efkkslasae,eEkls aaad.

a. CO4 **Simulate** the Huffman Encoding algorithm to encode the message. Construct Huffman Tree, generate the codeword for each character, show the encoded message and count the total number of bits required to store the message. **7**

b. CO5 **Explain** the advantage of Huffman Encoding over the ascii codes approach mathematically. **1**

c. CO4 After reading the message Bob replied to Alice by sending an encoded message: 100010101011111101010. **2**
Now **use** the Huffman Tree constructed in question 2(a) to decode the message.

2.

Alice and Bob both are the peers of a decentralized distributed network. Alice wants to send a message to Bob using the platform but she does not want that any other users of the network would be able to understand the message. That is why she encodes the message using an encoding algorithm which is known to only both of them and sends it to Bob. The message is:

Saaees eklassddd,Aeeesss lkes.

a. CO4 **Simulate** the Huffman Encoding algorithm to encode the message. Construct Huffman Tree, generate the codeword for each character, show the encoded message and count the total number of bits required to store the message.

b. CO5 **Explain** the advantage of Huffman Encoding over the ascii codes approach mathematically.

c. CO4 After reading the message Bob replied to Alice by sending an encoded message: 1111110000111011010.
Now **use** the Huffman Tree constructed in question 2(a) to decode the message.

3.

a. CO2 You are given the following table containing symbols and their frequencies:

Symbol	A	B	C	D	+
Frequency	40	10	20	15	15

i) **Build** the Huffman code tree and find the codewords for each character.
ii) **Decode** 100010111001010 using the Huffman code that you generated.

4.

You are given the arrival and departure times of eight trains for a railway platform, each in the following format: $[arrival\ time,\ departure\ time)$. Only one train can use the platform at a time.

Suppose, you have got the following train-use requests for the next day.

$$\{ [8, 13),\ [6, 9),\ [11, 14),\ [2, 7),\ [1, 7),\ [12, 20),\ [7, 13),\ [13, 20) \}$$

i) **Find** the maximum number of trains that can use the platform without any collision. **03**
ii) **Determine** the minimum number of platforms that needs to be there to ensure the arrival and departure of all these trains without collision. **02**

5.

Following are the codes generated from a text for a Huffman tree construction.

H - 1000 *o* - 1001 <*space*> - 1010 *S* - 1011	*u* - 000 *d* - 001 *n* - 010	*e* - 011 *l* - 110 *t* - 111

You are also given the following information:

- The frequency of each leaf node except *e*, *l*, and *t* is 1.
- The left and right child nodes of the root have frequencies 5 and 8 respectively.

Now answer the following questions.

a) Suppose in a Huffman tree, the distances from the root to the pair of leaves denoting the letters *k* and *b* are 5 and 2 respectively, which letter between them is more frequent in the original text? Just **mention** the letter. **01**
b) **Draw** the Huffman tree from the given coding table above. **03**
c) Continuing on Q(b), what are the frequencies of *l*, and *t* in the original text? Just **mention** the frequencies. **02**

Your lazy friend, Tom, used a very simple encoding method for compressing a text file. The text file contains the string : *Sudden Hello*. He used a constant number of bits for encoding each of the ***n*** distinct characters in this text. The constant is $ceil(log_2 n)$ in this case. On the other hand, you have compressed the text file using your own huffman tree from Q(b).

d) **Mention** how many bits each of the characters in Tom's encoding scheme contain. **01**
e) **Compare** between the number of bits needed to decode the above-mentioned string using your friend's scheme and your Huffman tree from Q(b). **03**

6.

Following are the codes generated from a text for a Huffman tree construction.

J - 1000 *o* - 1001 <*space*> - 1010 *W* - 1011	*n* - 000 *h* - 001 *m* - 010	*l* - 011 *e* - 110 *c* - 111

You are also given the following information:

- The frequency of each leaf node except *e*, *l*, and *c* is 1.
- The left and right child nodes of the root have frequencies 5 and 8 respectively.

Now answer the following questions.

CO2

a) Suppose in a Huffman tree, the distances from the root to the pair of leaves denoting the letters *m* and *e* are 3 and 5 respectively, which letter between them is more frequent in the original text? Just **mention** the letter.
b) **Draw** the Huffman tree from the given coding table above.
c) Continuing on Q(b), what are the frequencies of *e*, and *c* in the original text? Just **mention** the frequencies.

Your lazy friend, Tom, used a very simple encoding method for compressing a text file. The text file contains the string : *Welcome John*. He used a constant number of bits for encoding each of the ***n*** distinct characters in this text. The constant is $ceil(log_2 n)$ in this case. On the other hand, you have compressed the text file using your own huffman tree from Q(b).

d) **Mention** how many bits each of the characters in Tom's encoding scheme contain.
e) **Compare** between the number of bits needed to decode the above-mentioned string using your friend's scheme and your Huffman tree from Q(b).

7.

Consider the following modified ASCII scheme and the encoded message using it.

Character	a	b	c	d	e	f	g	h	i	j
m_ascii	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001

Message encoded with m_ascii code:
0000 0100 0100 1000 0100 1000 0100 1000 0001 0101 0001 0001 0110 0000 0100 1001 0010 1001

Now answer the following questions.

a) [CO1] **Decode** the message.
b) [CO1] **Show** simulation to find the encoding of each of the characters using Huffman coding technique. While creating a tree, less frequency nodes go to the left side of the root; left is considered as a 0, right is 1.
c) [CO1] **Calculate** the number of bits required in the encoded message using huffman technique.

8.

Consider the following modified ASCII scheme and the encoded message using it.

Character	a	b	c	d	e	f	g	h	i	j
m_ascii	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001

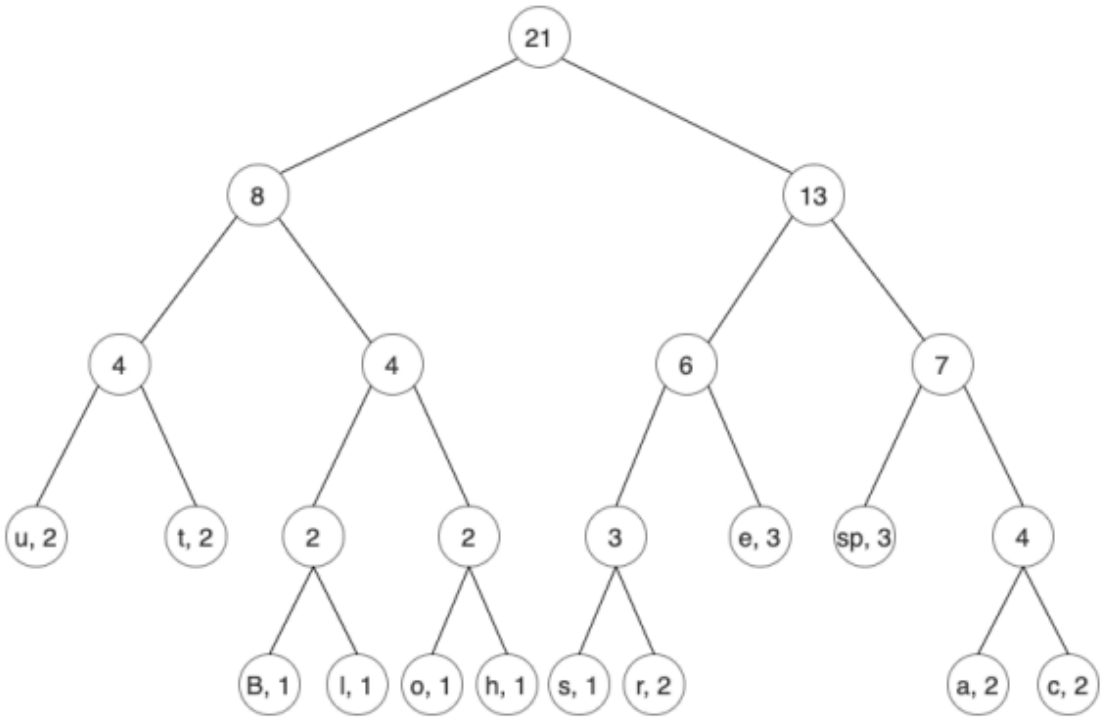
Message encoded with m_ascii code:
0010 0000 0010 0000 0011 0110 0111 1000 0011 1001 0000 0010 0011 0010 1001 1000 0010 0111

Now answer the following questions.

a) [CO1] **Decode** the message.
b) [CO1] **Show** simulation to find the encoding of each of the characters using Huffman coding technique. While creating a tree, less frequency nodes go to the left side of the root; left is considered as a 0, right is 1.
c) [CO1] **Calculate** the number of bits required in the encoded message using huffman technique.

9.

b) [CO1] You are given the following tree generated by simulating the Huffman encoding algorithm on a text message.



i) **Write** the encoding for each of the characters. Consider left to be 0, and right to be 1 when traversing the tree.

ii) Decode the following encoded message using the character codes you wrote on question (i).
0100100111101111000010111101100010110110001011110111010011011000111100 0101

10.

[CO1] Welcome to the exciting realm of Bracula Gaming World, where you can earn lots of money by playing different games. But there's a twist – time is limited to just 20 hours! Each game has its own time limit and prize money if played for the full duration. However, you can play for a **fraction of the time** and still earn money in proportion. Check the list of games below.
Your task is to make the most of your 20 hours and collect **as much money as possible**.

Game Name	**Total Time Limit**	**Prize Money For Full Duration**
Road Rush	16 hours	85$
GTA Vice City	20 hours	95$
Virtual Cop	4 hours	30$
House of the Dead	7 hours	45$
Far Cry	3 hours	18$
Devil May Cry	11 hours	50$

i) **Name** the algorithm tailored for this situation.

ii) **Write** the names of the games and their respective playing times that will lead to maximum prize money. Show a simulation of your algorithm.

Dynamic Programming

Today Alice has learned how to find the Longest Common Subsequence (LCS) of two given strings. Now she wants to find the LCS of “axyb” and “axyxb”. After hours of hard work, she has made this LCS table, M.

	empty	a	x	y	x	b
empty	0	0	0	0	0	0
a	0	1	1	1	1	1
x	0	1	2	2	2	2
y	0	1	2	3	3	3
b	0	1	2	3	3	4

a. **Explain** what you understand by the value M[3][4] = 3.
CO1

b. **Find** out the LCS String from the table. Show the steps of your work.
CO2

c. **Determine** the maximum profit for the 0-1 Knapsack problem given in the following table
CO4 using Dynamic Programming. Show the steps with a recursion tree or the memory matrix.

Knapsack Weight: 8 kg

Objects	***i***	***ii***	***iii***	***iv***
Weight (kg)	5	4	6	3
Profit ($)	9	8	20	9

2.

DNAs contain genetic and hereditary information of almost every living organism. DNA Sequence Matching is a procedure to compare two DNA sequences and find similarity between them. Given below is the recurrence relation of a dynamic programming algorithm to perform sequence matching of two DNAs.

$$seq(i,j) = \max \begin{cases} seq(i, j-1) - 1, \\ seq(i-1, j) + p(i, j), \\ seq(i-1, j) - 1 \end{cases}$$

where,

$$seq(i,j) = 0, \quad if\ i = 0\ or\ j = 0$$
$$p(i,j) = 1, \quad if\ DNA1[i] = DNA2[j]$$
$$p(i,j) = -2, \quad otherwise$$

As you can see, the formula takes two indices, i *and* j of two DNA sequences given as input. Then it produces a similarity score as output. We want to apply this formula to compare the following DNA sequences.

index	1	2	3	4	5
DNA1	G	C	G	T	A
DNA2	C	T	G	A	G

Now answer the following questions:

a) [CO1] Put appropriate values to **fill the gaps** in the line below (answer in your script): **02**
$seq(__, __)$ *represents the final similarity score of the given DNAs.*

b) [CO1] **Apply** the formula to calculate a similarity score between the DNA sequences given above. Show your work with either a recursion tree or a memory table. **05**

c) [CO2] **State** the time complexity of this algorithm with proper reasoning. **02**
You can assume that, the lengths of the DNA sequences are N *and* M respectively.

d) [CO3] Mriaslun, a talented algorithmist, claims that he can implement the algorithm in an optimized way so that it only takes $O(min(N, M))$ space. Can you do it too? **Explain** your idea. **01**

3.

Two infamous thieves, Denver and Nairobi, planned to rob the famous Louvre Museum. Before the scene, they both agreed on the fact that none of them will break any item as all the items in the museum are too precious, and taking a fraction of any item won't sell on the black market. If it fits in the bag as a whole, they will take it, otherwise, leave it as it is.

Both of them arrived at the Royal Treasury with an empty knapsack weighing a total of **7 kg** each. Even though both thieves are experts in their fields, they take slightly different approaches. Denver believes he will use a Dynamic Programming Approach to rob the items in the most efficient manner possible. Nairobi, on the other hand, believes that if she chooses a Greedy Approach, she will make the most money.

The objects in the Royal Treasury Museum are listed below.

Objects	Diamond	Jewelry	Sculpture	Painting	Gold Crest
Profit ($)	3	4	12	9	12
Weight (kg)	1	2	8	4	5

a) **Simulate** your dynamic programming algorithm to find the maximum profit Denver can make.
b) From your memory/dp table, **find** out which items would he take to make this amount of profit?
c) Does Nairobi's belief remain valid after the robbery? **Prove** it.

4.

You are Monco, a video game character, you have just stepped into level 2. This game requires you to acquire necessity items to sustain through a level and this is done by trading reward points from the previous level. From level 1 you have gained 9,000 points. Below is a list of items you can get by trading your reward points. Each item has certain properties but be wise to pick the ones which will maximize your sustainability by maximizing mission impact.

	Gold Bar	Pain Killer	Waterproof Boots	Sun glasses	Energy Bar	Night Vision
Trade Price (points)	8000	2000	2000	2000	1000	3000
Mission Impact	1	5	3	2	2	4

a) **Simulate** your chosen algorithm on the below items to find your picked items.
b) If you were to apply brute force, what would be the time complexity and why? **Explain.** Assume that points you have gained are denoted as P, and the number of items is N.

5.

Wye is a CSE BracU Student. The pre-advising week is approaching, and he is trying to find the most suitable course combination for next semester. As per his previous academic results, his credit limit **has been set to 9**. Different courses have different credits. He came up with values for these courses by consulting with students who have already taken them. He wants to select the most valued courses.

Courses, credits and values are provided below:

Course	PHY112	CSE330	CSE331	CSE370	CSE499
Credit	1	2	3	4	5
Value	10	8	16	13	17

a. Wye wants to try all possible combinations to find the best one. Help him by simulating a suitable dynamic programming approach. **Show** your work in detail, then **write** down the selected courses.
b. Zed, a friend of Wye, adopted a slightly different approach and considered the courses based on their per-credit value. **Determine** whether Zed could select more valued courses than Wye. Note that, selecting a course still means taking the whole, with full credits.
c. If Zed is successful, he will send a message **Successful**; otherwise, **Unsuccessful**. Instead of fixed-length coding, he is willing to send the message using Huffman Coding. **Show** simulation of Huffman coding to encode his message.